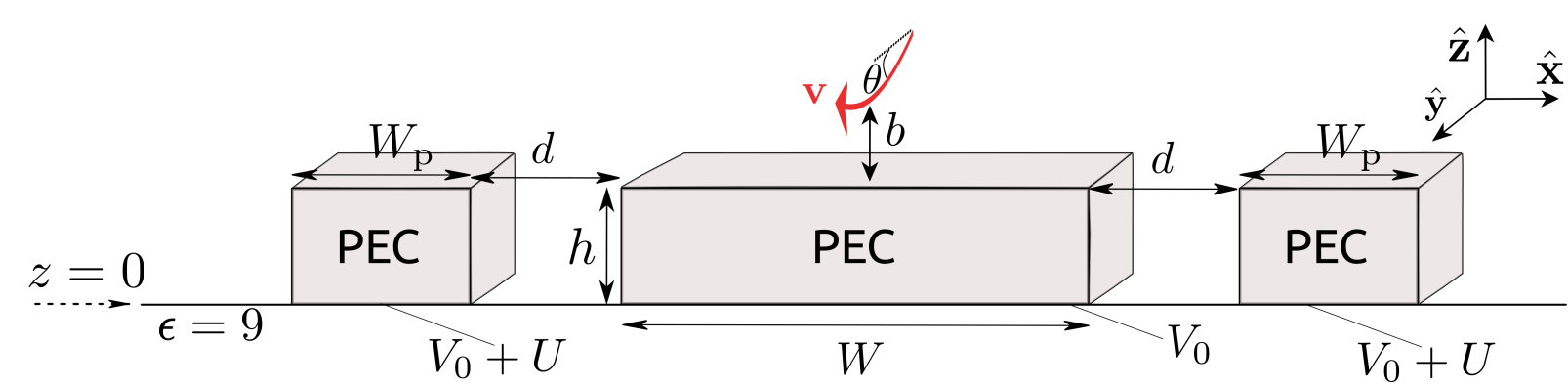


Electron-coupling to waveguide: new potential



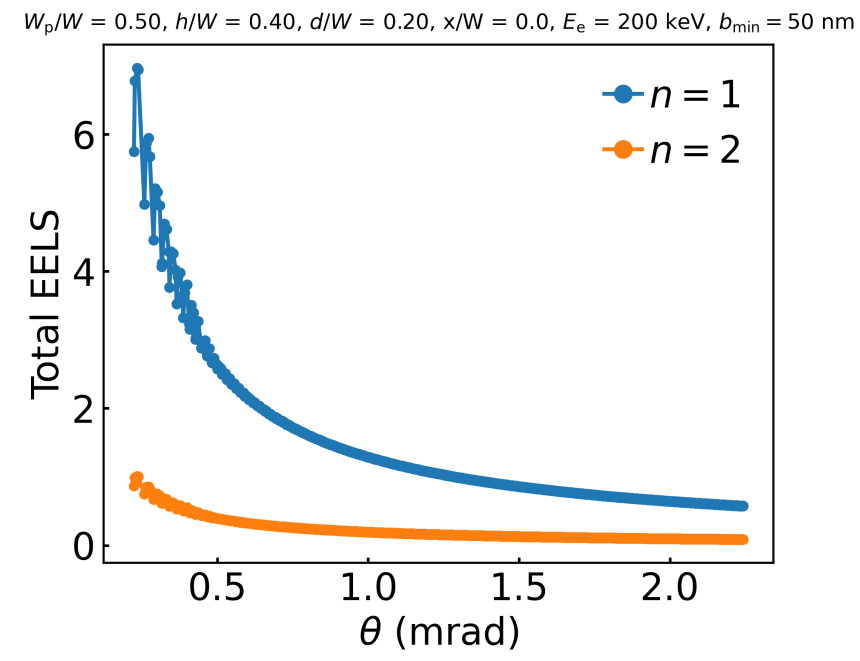
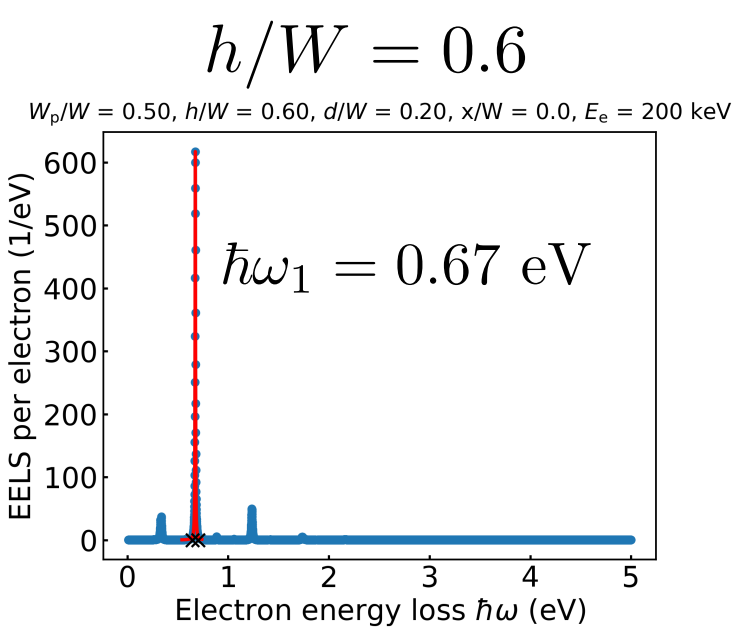
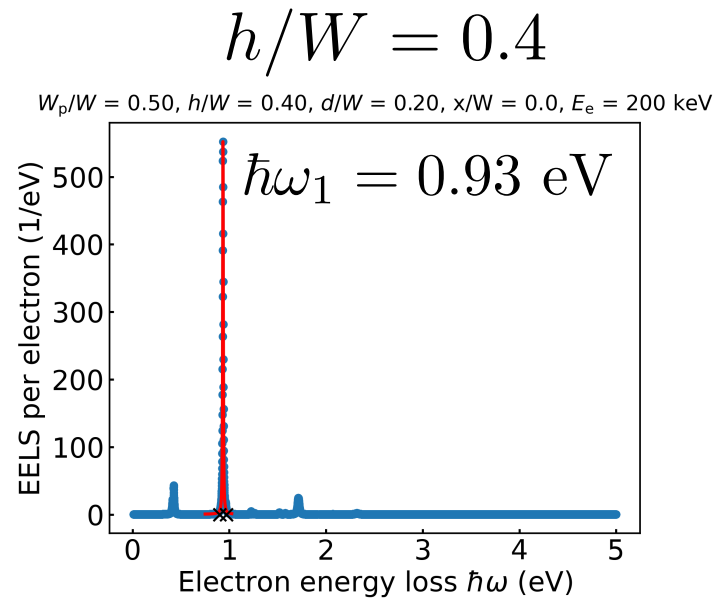
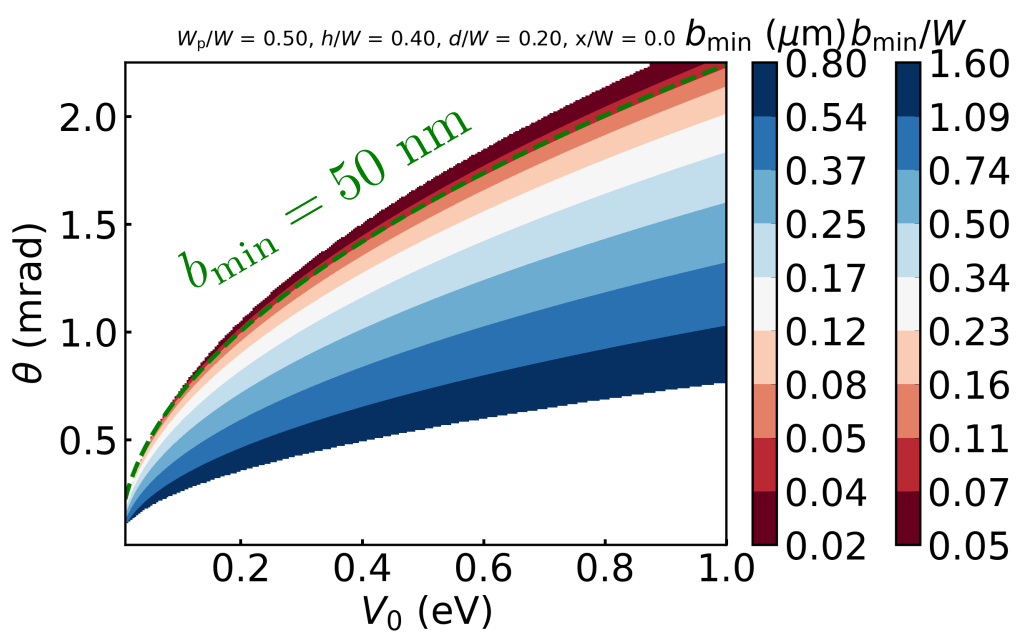
$E_e = 200 \text{ keV}$
 $W_p/W = 0.5$
 $d/W = 0.2$
 $x/W = 0$

Correction of the trajectory:

$$0 = \frac{V(z_{\min})}{U} \frac{U}{V_0} + \frac{m_e c^2 \gamma_e}{2e} \frac{\beta^2 \sin^2 \theta}{V_0}$$

$\downarrow V(|x| \leq 0.5, z = h) = V_0/U$

$\longrightarrow \Gamma_{\text{EELS}}(\omega) = 2 \int_{b_{\min}+h}^{\infty} \frac{v dz}{\sqrt{\left| c^2 \beta^2 \sin^2 \theta + \frac{2eV(x, z)}{m_e \gamma_e} \right|}} \frac{d\Gamma_{\text{EELS}}(\omega, x, z)}{dy}$



(running: EELS~5)